

Zamson Lim Zhen Pin

Cybersecurity — SOC Engineering, Detection & Purple Team

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PROFILE

Final-year BSc (Hons) Cybersecurity student at Asia Pacific University. After a ransomware incident left my employer with no way to reconstruct what had happened, I proposed a Security Operations Centre, got approval from the Head of IT, and **built and ran it as the only person on security** — architecture, detection engineering, incident response, purple-team exercises and reporting, all self-owned. Comfortable working from the raw data rather than from a vendor dashboard, and in the habit of verifying against the system itself instead of the message it reports.

EXPERIENCE

Network & Security Administrator (Intern)

Aug 2025 – Aug 2026

Palmgold Management Sdn Bhd · Kuala Lumpur · sole owner of security operations

- Proposed and built the company's **Security Operations Centre** from nothing after an August 2025 ransomware incident, growing it from a proof of concept on a spare desktop to the production stack now running the company's monitoring.
- Deployed a **Wazuh SIEM with co-located OpenSearch** on a Proxmox host I set up myself, integrating firewall, core and access switching, wireless, Windows endpoint, Active Directory and Zeek network telemetry through custom ingest pipelines.
- Rolled out monitoring agents to **80+ Windows machines** via Active Directory Group Policy using an idempotent, safely re-runnable installer with per-machine logging.
- Built the network sensor: port mirroring from the core and access switches into a multi-node Zeek cluster through a Python decapsulation stage, giving visibility into internal traffic the firewall never sees.
- Wrote custom decoders and correlation rules matched to the hardware actually in use, deliberately without an endpoint-agent dependency I had no authority to deploy fleet-wide.
- Ran **authorized purple-team exercises** — attacking from Kali Purple, then detecting from the SIEM — and turned each into detection content; surfaced an unmonitored network segment that no rule could have covered.
- Built and analysed malware samples in isolated lab VMs, purely to validate and tune signatures.
- Ran an **authorized internal vulnerability assessment** and delivered three audience-tailored reports; honestly downgraded one finding to "unconfirmed" when it could not be reproduced.
- Built the SOC's own operator dashboards, served from the SIEM host behind a hardened reverse proxy.
- Recovered a full ingestion outage caused by archive logging filling the disk and silently flipping every index read-only; installed retention policies so it could not recur.
- Provided day-to-day IT support at head office and on site across branch locations, and handed over a documented, running SOC at the end of the internship.

OPEN-SOURCE TOOLING

Three single-file SOC consoles, de-identified and published

2026

zamsonl.github.io/portfolio-website · live demos and source

- **SOC Traffic Map** — a D3 world map over a firewall-fed alert index, with a threat-posture score that decays over time rather than accumulating, and rate buckets keyed off the event timestamp so a batched poll cannot invent a spike.
- **SOC Alert Console** — incremental polling and severity bucketing over a Wazuh index, with every filter routed through one dispatcher so counters, ranked lists and the feed cannot disagree; seven surface themes across six severity palettes, all contrast-checked.
- **Packet Highway** — a Three.js visualizer rendering firewall traffic as vehicles on a night highway, using instanced meshes and object pooling to hold frame rate on a live feed.
- Each is one HTML file with no build step, no dependencies to install, and zero external requests; every organisation name, hostname and address replaced with a placeholder before publication.

SELECTED PROJECTS

Zeek network monitoring pipeline

- Mirrored traffic decapsulated in Python and replayed onto a virtual interface into a multi-node Zeek cluster.
- Detections for domain-generation bursts, DNS tunnelling, exfiltration and inbound scans.
- Re-characterised an alarming reported capture loss as an MTU artefact of the mirror tunnel, which decides which detections can be trusted.

Malicious file analysis (Autopsy)

- Forensics exercise at APU examining a lecturer-provided sample in Autopsy.
- Worked through the file’s contents and metadata rather than trusting a scanner verdict; written up as a formal report.

Building with AI tooling

- Built and debugged much of the SOC tooling with Claude Code, and ran local models through Ollama to judge what genuinely needs a hosted one.
- The discipline is the point: what to delegate, what to verify, and never trusting a “done” message over the artefact itself.

Active Directory security monitoring

- Learned AD properly — domain structure, Group Policy, event log semantics — then built monitoring on native Windows event logs.
- Saved searches for authentication failure, log clearing and account enumeration, ranked by volume.

Wireless & access point monitoring

- Wireless syslog brought into the SIEM with custom decoders for association, authentication and roaming events.
- A scheduled enrichment pipeline resolving hardware addresses to hostnames, so alerts name a machine rather than a MAC.

University coursework

- **Cisco Packet Tracer labs** — VLANs trunked and routed across two switches, verified end to end.
- **Library Management System** (Python) — role-based CLI with attempt limiting and file-backed persistence.
- **Medical Management System** (Java) — inheritance and interfaces over a long branch statement.

SKILLS

SIEM	Wazuh (deployment, agent management, custom decoders and correlation rules), OpenSearch query DSL and aggregations, ingest pipeline design, file integrity monitoring, alert triage and incident response
NETWORK	Zeek network security monitoring, FortiGate policy and syslog integration, Aruba switching and wireless, traffic mirroring and decapsulation, Wireshark, Cisco Packet Tracer
SYSTEMS	Proxmox virtualization, Ubuntu / Rocky Linux administration, systemd tuning, Nginx reverse proxy and TLS, Windows Server, Active Directory and Group Policy at fleet scale
OFFENSIVE	Authorized vulnerability assessment (OpenVAS, Nmap, Nikto), purple-team exercise design and execution, malicious code analysis in isolated labs, Kali Linux
FORENSICS	Autopsy file and disk analysis, incident write-ups and handover documentation, multi-audience technical reporting
CODE	Python, JavaScript (D3.js, Three.js), Bash, Windows batch, Java, C++, C#, Rust
LANGUAGES	English, Mandarin, Malay, Cantonese

EDUCATION & COMPETITIONS

BSc (Hons) Cybersecurity

2024 – 2027

Asia Pacific University · final year

- Coursework across network security, digital forensics and secure development.
- Industrial internship completed at Palmgold Management, with the SOC build-out as the primary project.

Capture The Flag competitions

2024 – present

SekaiCTF · CYNX · BeSec · picoCTF

- **SekaiCTF 2025** with team Digital Phantoms — solved a medium reverse-engineering challenge on an Android APK using jadx-gui, keytool and Python, plus a cryptography challenge.
- Web exploitation, forensics and OSINT under time pressure; ongoing practice across categories.